5. a. limit aljabar

$L_1 : L_1 = \lim_{x \to k} \frac{x^2 - k^2}{x - k}$ faktorkan : $(x-k)(x+k)/(x-k)$

$= x + k$

subtitusi $k = 9$. $= 9 + 9 = 18$

$L_1 = 18$

$L_2 : L_2 \lim_{x \to 0} \frac{\sqrt{x(k+1)^2} - (k+1)}{x}$

$k = 9 \rightarrow (k+1) = 10$

sekawan : $\frac{x}{x\sqrt{x} + 100 + 10} = \frac{1}{\sqrt{x} + 100 + 10} = \frac{1}{20}$

$L_2 = 1/20$

$L_3 : L_3 \lim_{x \to 100} \frac{(a+2)x^3 + bx + 2}{x^3 - x + k}$

subs : $a = 2$, $b = 6$ $\rightarrow (a+2) = 4$

$= \frac{4x^3 + 6x + 2}{x^3 - x + 9}$

ambil pangkat tinggi

$L_3 = 4$

$= 4$

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5c. limit trigonometri

$L\ 4 = \lim_{x \to 0} = \frac{\sin((a+1)x)\tan((b+2)x)}{x^2}$

subs = $(a+1) = 3$
$(b+2 = 8$

$= \frac{\sin(3x)\tan(8x)}{x^2}$

gunakan
$\sin(kx) = kx$
$\tan(kx) = kx$

$= \frac{(3x)(8x)}{x^2} = 24$

$4\ 4 = 24$

teorema : $\sin x / x$

5d. limit clu

$(a+b+1) = 2+6+1 = 9$

$R(n) = \frac{100n + k}{n + (a+b+1)} = \frac{100n + 9}{n + 9}$

$= 100$

limit = 100
interpretasi : stabil

5e. limit akar

sekawan → hasil :

$\lim_{n \to \infty} (\sqrt{n^2+9n} - \sqrt{n^2+6n}) = \frac{9-6}{2} = \frac{3}{2}$

5f kekontinuan

limit : $C(x) = \frac{x^2 - 4^2}{x - 4}$; $x < 4$; $x + 4 \to 2k = 18$
$p = 18$

syarat kanan : $ax + q$
harus : $ak + q = 2k$
$2(9) + q = 18$

$q = 0$

$p = 18$
$q = 0$

deli得力

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